

Economic Comparison of Wind Power and Gas-Powered Thermal Power Plants in Europe

Presented By:

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Thesis Statement:

Wind energy is cheaper than gas-fired thermal power plants.

Abstract:

Wind energy requires a higher initial capital investment of €1.35 million per MW but avoids fuel costs, leading to predictable financial outlays. Gas-fired power plants have upfront costs of €1.2 million per MW and incur fuel expenses of €0.07 per kWh. Over a 20-year lifespan, wind energy ensures stability through predictable O&M costs, while gas plants face fluctuating fuel prices influenced by geopolitical factors. Wind energy produces electricity at €0.1209/kWh, while gas-fired plants generate at €0.0905/kWh. The 2020 data pre-dates geopolitical shifts that have increased gas prices, suggesting that gas-powered thermal power plants may now produce electricity at a price that matches or exceeds wind power costs.

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